

Questions			Answers	Marks
8	(a)		Electrons have <u>zero potential energy at infinity</u>, and so less than this near the <u>positively</u> charged nucleus. Also accepted: Electrons are <u>attracted to the positively charged nucleus</u>. So work has to be done / energy has to be given to <u>remove the electrons from the nucleus</u> (where potential energy is zero) .	B1
	(b)	(i)	$E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{590 \times 10^{-9}}$	M1
			$E = 3.37 \times 10^{-19} = 3.4 \times 10^{-19} \text{ J}$	A1
		(ii)	electron transit from $-5.8 \times 10^{-19} \text{ J}$ to $-2.4 \times 10^{-19} \text{ J}$	B1
			absorbing <u>all</u> the energy of the incident photon	B1
	(c)		Incident light is directional; light is re-emitted in all directions.	B1

9	(a)	Fig. 9.2 shows an exponential curve and hence count rate tends towards zero but
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